sandwiched

Relations between EOT and OT

- **Limiting behavior:** $\mathrm{OT}_\varepsilon \xrightarrow{\varepsilon \to 0} \mathrm{OT}$
- **Bounds:** $\mathrm{OT} - \epsilon H(\pi_\varepsilon^\star) \leq \mathrm{OT}_\varepsilon \leq \mathrm{OT}$
  - Proof Upper Bound: Evaluate the regularized objective at any unregularized optimizer π^*:
    $\mathrm{OT}_\varepsilon \;\leq\; \langle C, \pi^\star \rangle \;-\; \varepsilon H(\pi^\star) \;\leq\; \langle C, \pi^\star \rangle \;=\; \mathrm{OT} \quad$ (since $H(\pi^\star) \geq 0)\,.$
  - Proof Lower Bound: By definition of π_ϵ,
    $\mathrm{OT}_\varepsilon \;=\; \langle C, \pi_\varepsilon \rangle \;-\; \varepsilon H(\pi_\varepsilon) \;\geq\; \min_{\pi \in \Pi(\alpha,\beta)} \langle C, \pi \rangle \;-\; \varepsilon H(\pi_\varepsilon) \;=\; \mathrm{OT} \;-\; \varepsilon H(\pi_\varepsilon)\,.$
- **Exercise:** provide a lower bound that does not depend on $\pi_\varepsilon^\star$
- **Takeaway:** EOT **sandwiched** between OT and **a** $\epsilon H(\pi_\varepsilon^\star)$ deviation from it

Sinkhorn Divergence Motivation

Credit: Gabriel Peyre